# BIT1123/BISE2093/DIT1113

# BIT1123 OBJECT ORIENTED PROGRAMMING

# ASSIGNMENT 1 - INDIVIDUAL

| No | Name | Student ID | Programme |
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| SUBJECT CODE | BIT1123 |
| PASSPORT NO | A03691443 |
| SUBJECT NAME | Object Oriented Programming |
| NAME OF LECTURER | NAZMIRUL IZZAD BIN NASIR |

## 1. Introduction:

At the start of this course I was beginning Java programming from zero. I had little experience with Java so the first few lessons were mostly about getting to know the language and understanding how Java programs are built.

As the lessons went from Week 1 to Week 10 I was introduced to object-orientated programming ideas and gradually started to understand how they are used in Java programs. The lessons gave me a chance to practice these ideas of just learning them in theory.

This assignment allowed me to bring together my workk from the lessons into one GitHub repository. It also helped me see my progress from writing Java programs to working with object-oriented ideas, collections, file handling and a graphical user interface.

## 2. Knowledge Gained from the Lessons

**Week 1**

Week 1 was my starting point with Java. Since I was learning Java from zero I focused on undrstanding the structure of a Java program and how to run simple programs. This gave me a base for the topics that followed in the lessons.

**Week 2**

In Week 2 I learned about classes and objects. This was one of the important steps in understanding Object-Oriented Programming.

I learned that a class can be used as a structure for creating objects and that methods can describe the behavior of those objects. Working with a Student example helped me understand the connection between a class and an object clearly.

**Week 3–4**

The Week 3–4 lessons introduced inheritance. Related OOP ideas.

I learned that a class can take properies and behaviors from another class. Working with classes such as Person, Student and Lecturer helped me understand how classes can be connected to each other of writing separate code for every type of object.

**Week 5**

Week 5 introduced encapsulation.

I learned about using attributes and accessing them through methods like getters and setters. At first this seemed unnecessary because it added code but I gradually saw that encapsulation provides better control over the data inside an object.

**Week 6**

Week 6 continued working with inheritance through an Employee and Lecturer example.

This helped me get better at understanding inheritance and constructors. I also became more familiar with how information from a parent class can be used by a child class.

Week 7

Week 7 introduced abstraction.

The appliance example helped me understand how an abstract class can define functionality while allowing individual classes to provide their own specific behavior. This was useful because I could see how OOP ideas can be used to organize a program than simply putting everything into one class.

**Week 8–9**

Weeks 8–9 introduced ArrayList and file handling.

I learned how an ArrayList can be used to store items and how information can be written to and read from a file. File handling was one of the diffiult areas for me because it was different from the basic Java programs I had worked with earlier. I had to understand how files are opened, written to and read from while also dealing with errors.

**Week 10**

Week 10 introduced GUI programming using Java Swing and event handling.

This was another topic for me because GUI programming involved several components working together. Of simply displaying output in the console the program had to respond to user actions such as clicking buttons. Although it was difficult at first it gave me an idea of how Java can be used to create interactive applications.

## 3. Challenges Faced

The biggest challenge throughout these lessons was that I was learning Java from the beginning.

At first basic Java syntax and the structure of a program required attention. As the lessons became more advanced I also had to understand new ideas at the same time. I found classes and objects, inheritance, encapsulation, abstraction and ArrayList challenging while I was learning them. These ideas are connected to each other so understanding one idea properly was important before moving on to the one.

The two topics I found difficult were file handling and GUI development.

With file handling I had to understand how Java interacts with files and how exceptions can happen during these operations. GUI develoment was also challenging because there were components involved and the program needed to respond to user interaction.

I also found GitHub confusing at first. I was not familiar with creating repositories organizing folders and uploading files correctly. During this assignment I had to learn how to structure my repository and make sure that each lesson was placed in its folder.

## 4. How I Overcame the Challenges

I overcame these difficulties by taking the ideas step by step instead of trying to understand everything at once.

For the OOP ideas I worked with examples and focused on understanding what each class, object and method was doing. Once I understood the idea it became easier to see how inheritance, encapsulation and abstraction were connected.

For file handling and GUI programming I had to spend time understanding the structure of the code and how different parts worked together. Repeatedly going though the examples helped me become more comfortable with them.

I also learned through testing and correcting mistakes. When a program did not work as expected looking at the code and identifying where the problem occurred helped me understand the programming ideas better.

For GitHub I learned by creating the repository and organizing the assignment step by step. This helped me become more comfortable with repository structure, folders and uploading files.

## 5. Improvement in My Java Programming Skills

At the beginning of the lessons I was starting Java from scratch. By the end of Week 10 I felt that I had developed an understanding of Java and could write simple Java programs.

I became comfortable creating classes, objects, constructors and methods. I also gained experience working with inheritance, encapsulation, abstraction and ArrayList.

Another improvement was my ability to read and understand Java code. Initially code containing classes could be confusing. After completing the lessons I became better at identifying what each class and method was responsible for.

I also gained some experience with file handling and GUI programming even though these remained more challenging areas for me.

## 6. Understanding of Object-Oriented Programming

The lessons helped me understand that Object-Oriented Programming is about organizing a program around objects and the classes that define them.

I now have an understanding of several important OOP ideas:

Classes and objects. A class provides the structure from which objects can be created.

Inheritance. Allows a class to take charactristics and behavior from another class.

Encapsulation. Controls access to data inside a class.

Abstraction. Focuses on functionality while hiding unnecessary implementation details.

Polymorphism. Allows objects to behave differently through shared methods.

Before completing these lessons these terms were mostly unfamiliar to me. Now I can recognize them in Java programs. Understand their basic purpose.

## 7. Future Learning Goals

Although I now understand the ideas and can write simple Java programs I still have a lot to learn. My next goal is to become more confident with Java by practicing regularly instead of only workig on lesson exercises. I would like to improve in file handling, GUI development and writing larger programs.

I also want to strengthen my understanding of OOP by building projects where several ideas are used together. This would help me mov from understanding individual ideas to applying them naturally when developing a program. In the future I would also like to learn advanced Java programming and use these programming skills in cybersecurity-related projects.

## 8. Conclusion

Overall this assignment was a learning experience because I started the lessons with very little Java knowledge and gradually developed a basic understanding of the language and Object-Oriented Programming.

The important thing I gained was not only knowledge of individual Java ideas but also an understanding of how those ideas can be combined to create organized programs.

I still have areas that I need to improve, particularly file handling and GUI development. I now feel more confident about continuing to learn Java. The lessons gave me a founation that I can build on through practice and programming projects.

Creating the GitHub repository also gave me experience, in organizing programming work and maintaining a structured collection of my lesson projects.

## 9. GitHub Repository

**Repository: ArmanGani_BIT1123-OBJECT-ORIENTED-PROGRAMMING**

**GitHub URL: https://github.com/armanganiofficial-git/ArmanGani_BIT1123-OBJECT-ORIENTED-PROGRAMMING/tree/main**